GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

214139.1.pdf *Geochronology Record 1906*

214139.1: monzogranite gneiss, Joker Well

(Goonetarra Granodiorite, Rothsay Suite, Annean Supersuite; Youanmi Terrane, Yilgarn Craton)

IOH Fielding, Y Lu, MTD Wingate and I Zibra

Location and sampling

YALGOO SH50-02, YALGOO 2241 Zone 50, MGA 474451E 6850870N
Warox Site AETYLG000113 Sampled on 24 May 2014

This sample was collected from a boulder-style outcrop on Bunnawarra Station, about 22.5 km northeast of Bunnawarra Homestead, 15.2 km northwest of Badja Homestead, and 4.8 km southwest of Joker Well.

Geological context

The unit sampled is the Goonetarra Granodiorite of the Rothsay Suite, Annean Supersuite (Van Kranendonk et al., 2013; Zibra, 2020; ENS Report Goonetarra Granodiorite, viewed 28 April 2023). The 2763–2749 Ma Goonetarra Granodiorite consists mainly of metatonalite, metagranodiorite and metamonzogranite, which are typically foliated and lineated (Zibra, 2020: ENS Goonetarra Granodiorite, viewed 28 April 2023). In the Yalgoo dome, the Goonetarra Granodiorite is intruded into the Kynea Tonalite and into the greenstone succession (Zibra et al., 2018; Clos et al., 2018). Near the centre of the Yalgoo dome, the Goonetarra Granodiorite has intruded by monzogranite of the 2650–2600 Ma Walganna Suite of the Bald Rock Supersuite (Zibra, 2020: ENS Report Goonetarra Granodiorite, viewed 28 April 2023). A granodiorite gneiss and monzogranite gneiss of the Goonterra Granodiorite sampled about 90 m to the southwest and 1.5 km to the east-southeast yielded crystallization ages of c. 2763 Ma (GSWA 214138, Fielding et al., 2023b; GSWA 214101, Lu et al., 2017).

Petrographic description

The sample is a massive to weakly foliated, medium-grained monzogranite gneiss, consisting of about 35–40% plagioclase, 30–35% K-feldspar, 25% quartz, 5% biotite, 1–2% opaque minerals, 1% myrmekite, and accessory apatite, allanite and zircon. Plagioclase is anhedral, equant to tabular, up to 3.5 mm in length and poorly twinned. Some grains exhibit a distinct core marked by very fine sericite. K-feldspar occurs as anhedral, equant to slightly elongate grains that are up to 4 mm long. Many have very fine microperthitic texture and microcline twinning, and some of the larger grains contain small inclusions of plagioclase, quartz and biotite. K-feldspar porphyroclasts are wrapped by elongate trails of biotite and quartz ribbons, defining the metamorphic foliation. Strain-induced myrmekites (<0.5 mm) commonly occur along K-feldspar margins parallel to the foliation. Quartz is weakly to moderately strained and occurs as irregular aggregates intergrown with plagioclase and K-feldspar. Finer grained aggregates of quartz and microcline have a polygonal texture. Biotite is partially replaced by chlorite and epidote and occurs as subhedral to anhedral laths, up to 1.5 mm long. Subhedral opaque grains up to 0.7 mm in diameter are probable magnetite. The accessory minerals include subhedral, equant to tabular apatite up to 0.15 mm, metamict allanite up to 0.25 mm in diameter and dark brown zircon that are located within biotite. Microstructures indicate that the main fabric developed under mid- to low-amphibolite facies conditions.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and subhedral to euhedral. The crystals are up to 500 µm long, and elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. Many crystals contain high uranium, metamict zones. A CL image of representative zircons is shown in Figure 1.

Analytical details

This sample was analysed on 18–19 May 2015, using SHRIMP-B. Ten analyses of the BR266 standard were obtained during the session. Significant drift of standard $^{238}U/^{206}Pb^*$ dates during the session was addressed by fitting a LOWESS curve (Cleveland, 1979) with a smoothing window of six analyses, implemented using the program Isoplot 2.50 (Ludwig, 2009; Fielding et al., 2023a). During the session, ten standard analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.10% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

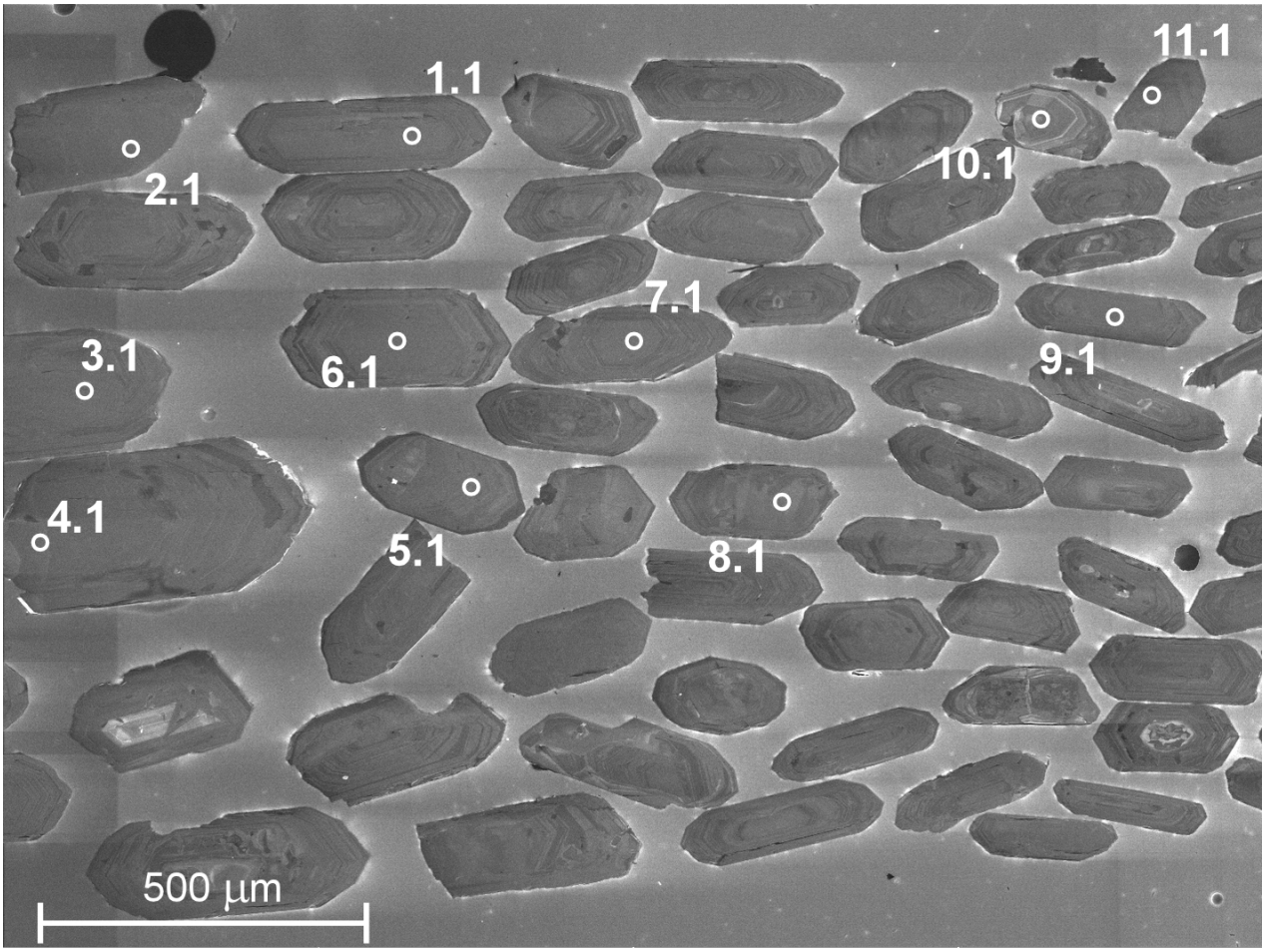

Figure 1. Cathodoluminescence image of representative zircons from sample 214139: monzogranite gneiss, Joker Well. Numbered circles indicate the approximate locations of analysis sites

Results

Twelve analyses were obtained from 12 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 2).

Interpretation

The analyses are concordant to moderately discordant (Fig. 2), and form two groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Group I comprises ten analyses of ten zircons (Table 1), which yield a weighted mean $^{207}Pb^{*}/^{206}Pb^{**}$ date of 2761 ± 4 Ma (MSWD = 0.57).

Group P comprises two analyses of two zircons (Table 1), which yield $^{207}Pb^{*}/^{206}Pb^{*}$ dates of 2741 ± 6 Ma and 2747 ± 4 Ma (1σ).

The date of 2761 ± 4 Ma for the ten analyses in Group I is interpreted as the magmatic crystallization age of the monzogranite protolith.

The dates of 2741 and 2747 Ma for the two analyses in Group P are interpreted to reflect minor ancient loss of radiogenic Pb.

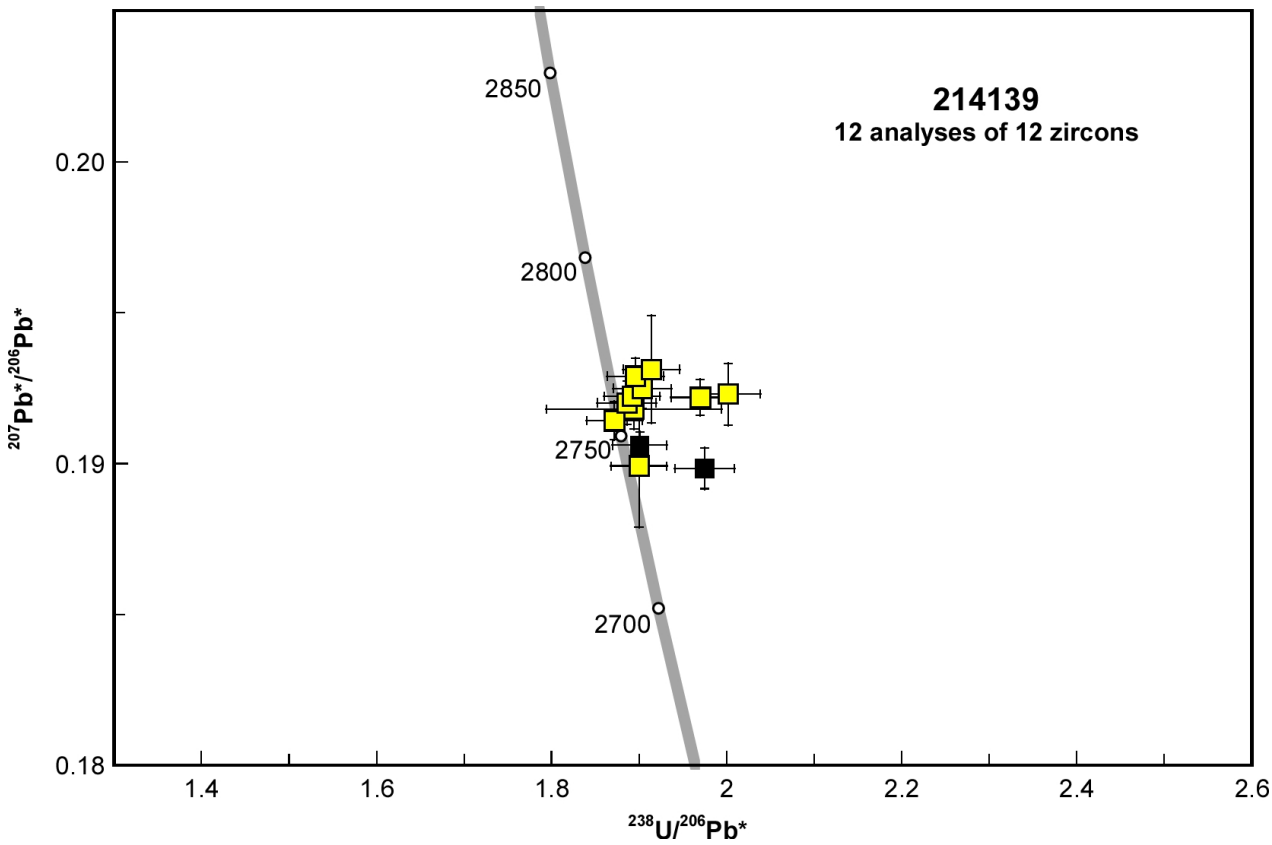

Figure 2. U–Pb analytical data for zircons from sample 214139: monzogranite gneiss, Joker Well. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss)

Links

Table 1. Ion microprobe analytical results for zircons from sample 214139: monzogranite gneiss, Joker Well: 214139.1.Table1.xls
SHRIMP spot images: 214139_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

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